Mid-Semester Examination, October 2026

Course: Discrete Mathematics. Time: 90 minutes. Maximum marks: 30. Answer all questions.

• Q1 (Unit 1). Prove that $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$. [5]

• Q2 (Unit 1). Show that congruence modulo 5 is an equivalence relation on $\mathbb{Z}$. [5]

• Q3 (Unit 2). Using a truth table, check whether $(p \to q) \wedge (q \to r) \to (p \to r)$ is a tautology. [5]

• Q4 (Unit 2). Negate: 'Every student in the class has a laptop or a tablet.' [5]

• Q5 (Unit 3). In how many ways can 8 people sit around a round table? [5]

• Q6 (Unit 3). Prove that among any 13 people, two were born in the same month. [5]